Velocity or salinity? Ocean reanalyses disagree on how the Atlantic is freshening, and CMIP6 models split the same way

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Abstract

The Atlantic meridional overturning circulation (AMOC) is widely anticipated to weaken under sustained greenhouse forcing, but the direct observational record remains too short (∼20 years at 26.5°N from the RAPID array) to distinguish a forced trend from internal variability. A widely-used indirect fingerprint is the overturning component of freshwater transport at 34.5°S, F_{ovS}, whose sign identifies the salt-advection feedback regime. Four independent ocean reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4), all analysed here with a consistent implementation of the canonical de Vries & Weber decomposition that *subtracts* the section-mean (barotropic) velocity to isolate the overturning component, agree that F_{ovS} is negative on the long-term mean (-0.02 to -0.12 Sv), placing the Atlantic in the bistable regime. But they disagree along multiple axes when the record is examined in more detail: two products (ORAS5, GLORYS12V1) show a significant declining trend while two (SODA, ECCO) show a flat or non-significant trend, and decomposing the change in F_{ovS} between a 1993–2005 baseline and a 2013–2025 recent window reveals fundamentally incompatible mechanism attributions: ORAS5 attributes 89% of its decline to circulation-structure change, GLORYS12V1 attributes 87% to salinity redistribution—the direct theoretical signature of the salt-advection feedback—and SODA 3.15.2 is mixed. In CMIP6 (15 models, 12 with forced weakening), the mechanism split (6 salinity-dominant, 5 velocity-dominant, 1 mixed) mirrors this observational spread, demonstrating it reflects genuine physical diversity rather than numerical artefact. Partitioning CMIP6 AMOC projections by mechanism recovers the headline estimate of recent observational-constraint studies [Portmann et al., 2026] only in the salinity-dominant subset (median 52% AMOC weakening by 2100), while the velocity-dominant subset projects only 42%. The mechanism choice therefore imposes a near-10-percentage-point uncertainty on the near-term AMOC trajectory that is not resolved by current observational-constraint frameworks. We further show that the reanalysis-level mechanism disagreement cannot be explained by the Argo observing-system shock alone: repeating the decomposition on fully-post-Argo windows (Supplementary Fig. S6) preserves the product

split. Targeted in-situ observations of Atlantic interior salinity structure at 34.5°S remain required to close the residual gap.

Significance: We show that the observational-constraint chain linking surface reanalyses to future AMOC projections depends not only on the magnitude of the freshwater-transport decline, but on whether that decline is mostly velocity-driven or mostly salinity-driven—and those two interpretations imply nearly 10 percentage points of difference in the AMOC weakening expected by 2100. The mechanism disagreement is robust to observing-system-shock corrections and to the canonical barotropic-subtraction convention of the F_{ovS} definition.

Main text

Introduction

The Atlantic meridional overturning circulation (AMOC) transports heat from the subtropics into the North Atlantic and sustains the mild European climate. A sustained weakening of the AMOC would cool Europe by several degrees, amplify sea-level rise along the US East Coast, shift the tropical rainfall belt southward, and perturb marine ecosystems across the basin. Quantifying the magnitude and timing of any AMOC weakening is therefore one of the most pressing questions in climate science. The direct observational record is short: the RAPID-MOCHA array at 26.5°N has monitored the overturning only since 2004, and statistical techniques have struggled to distinguish a forced trend from natural multidecadal variability in twenty years of data.

In the absence of a long direct record, indirect *fingerprints* are used to diagnose AMOC state. The theoretical literature identifies the overturning component of freshwater transport at 34.5°S, F_{ovS}, as especially informative: its sign distinguishes the bistable from the monostable regime of the salt-advection feedback that underlies long-standing models of AMOC tipping [Rahmstorf, 1996, de Vries and Weber, 2005, Dijkstra, 2007, van Westen et al., 2024]. A negative F_{ovS} indicates that the overturning exports freshwater *from* the Atlantic, so that a weakening AMOC would further concentrate salt in the North Atlantic — a positive feedback that can, under sufficient forcing, drive the system to a collapsed equilibrium. A positive F_{ovS} indicates the opposite regime, with no such self-amplifying mechanism.

Recent studies converge on the conclusion that the Atlantic is in the bistable regime. Dong and co-authors [Dong et al., 2024] used a combination of XBT transects, Argo-based products, numerical ocean models, and coupled climate models and reported a mean F_{ovS} of -0.15 ± 0.09 Sv at 34.5°S. Portmann and colleagues [Portmann et al., 2026] developed a ridge-regularised emergent-constraint framework using 19 observable variables and concluded that the historical AMOC weakening, extrapolated to 2100 under SSP5-8.5, will reach $51 \pm 8\%$ — about 60% stronger than the CMIP6 multi-model mean. A common critique of the F_{ovS} fingerprint family remains, however: the observed decline may reflect wind-driven changes in the South Atlantic subpolar gyre rather than a buoyancy-driven salt-advection signature [Chemke et al., 2020]. If so, the fingerprint overestimates the tipping risk because wind-driven transport variations are transient and largely reversible.

Here we test the wind-versus-buoyancy hypothesis by decomposing the observed F_{ovS} decline into its velocity-driven and salinity-driven components. The decomposition is exact (Methods Eqn. 6) and can be applied independently to any reanalysis or climate-model section at 34.5°S. A velocity-dominated decline is *consistent with* wind-driven circulation change; a salinity-dominated decline is the direct theoretical signature of salt-advection feedback and cannot be explained by wind alone. We apply the decomposition to four independent ocean reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4) and to 15 CMIP6 coupled models. The central result is that two of the most widely-used high-quality reanalyses disagree not only in magnitude but in *mechanism*, and the ensuing uncertainty translates into a 14-percentage-point gap in AMOC weakening projected by 2100 when CMIP6 models are conditioned on mechanism class.

Results

Robust regime diagnosis across reanalyses (Fig. 1). All four reanalyses confirm a negative F_{ovS} at 34.5°S, placing the observational estimate firmly in the bistable regime [Rahmstorf, 1996, de Vries and Weber, 2005, Dijkstra, 2007, van Westen et al., 2024]. Mean values over their respective periods are: ORAS5 -0.024 Sv (1958–2025, $n = 68$); GLORYS12V1 -0.071 Sv (1993–2025, $n = 33$); SODA 3.15.2 -0.009 Sv (1980–2022, $n = 43$; see below for the fate of the pre-registered $|F_{\mathrm{ovS}}| > 0.3$ Sv outlier rule); ECCO-V4r4 -0.116 Sv (1992–2017, $n = 26$). ORAS5, GLORYS12V1 and ECCO all lie within the band implied by the direct-hydrography estimate of -0.15 ± 0.09 Sv from 49 XBT transects [Dong et al., 2024]; SODA 3.15.2 lies near zero, at the monostable end of the observational spread. Trends, computed under three independent methods (naive OLS, Santer et al. [2000] N_{eff}, GLS Prais-Winsten AR(1); full Table S1), disagree on direction: ORAS5 is declining at -1.4 ± 0.1 mSv yr^{-1} and GLORYS12V1 at -4.4 ± 1.1 mSv yr^{-1} (both significant under all three methods), while SODA 3.15.2 ($+0.6 \pm 0.4$ mSv yr^{-1}) and ECCO-V4r4 ($+0.09 \pm 0.3$ mSv yr^{-1}) show weakly positive, non-significant trends. Only two of the four products therefore support the Portmann et al. [2026] narrative of an observationally constrained decline.

Outlier handling. Under the *uncorrected* $\mathrm{F}_o vSformulatheSODA3.15.2annualmeanfor1998is$-$1.77\ Sv, anorderofmagnitudebeyondanyotheryearandphysicallyimplausibleat34.5S; adirectprobeofthe$ $dayfileshowsanomalouslylowsalinityvalues(13--17\ PSUinsomegridcellsvs.theregionalclimatology$ $-37\ PSU), consistentwithacorrupteduploadontheUMDmirror.Wepre-registeredathresholdof|\mathrm{F}_{\mathrm{ovS}}|$ 0.3 Sv to flag such artefacts. However, under the *barotropic-corrected* kernel adopted throughout this paper, the 1998 value collapses to -0.018 Sv and is no longer an outlier: the artefact was largely a uniform-salinity-bias $\times$ non-zero-net-transport coupling which the baroclinic decomposition inherently filters out. The pre-registered outlier rule therefore does not trigger for any SODA year, and all 43 annual values 1980–2022 are retained in the mean, trend, and decomposition. This is an informative side-effect of the fix: it provides an independent diagnostic that apparent data artefacts in DA-reanalysis-derived F_{ovS} can be inflated by the common but incorrect omission of the barotropic subtraction.

Reanalyses disagree on mechanism (Fig. 2). An exact decomposition of the period-to-period change in F_{ovS} into velocity-driven, salinity-driven, and cross-term components (Methods Eqn. 6) yields fundamentally incompatible attributions:

- **ORAS5**: $\Delta F_{\mathrm{ovS}} = -34$ mSv; $\Delta F_v = -30$ mSv (+89%), $\Delta F_s = -5$ mSv (+15%); class: *v-dominant.*

- **GLORYS12V1**: $\Delta F_{\mathrm{ovS}} = -91$ mSv; $\Delta F_v = -13$ mSv (+14%), $\Delta F_s = -80$ mSv (+87%); class: *s-dominant.*

- **SODA 3.15.2**: $\Delta F_{\text{ovS}} = +13$ mSv; $\Delta F_v = +6$ mSv (+45%), $\Delta F_s = +6$ mSv (+46%); class: *mixed.* $|\Delta F_{\text{ovS}}|$ is 2.0× the piControl internal-variability median but well below its 95th percentile, so the mechanism attribution is formally below the pre-registered signal-to-noise threshold (Supplementary Fig. S4). Notably the sign of ΔF_{ovS} *reverses* in the fully-post-Argo window (see Supplementary Fig. S6 and the Discussion), underscoring SODA's sensitivity to observing-system coverage.

- **ECCO-V4r4**: $\Delta F_{\text{ovS}} = -2$ mSv, below the pre-registered $|\Delta F_{\text{ovS}}| \geq 10$ mSv threshold; the mechanism is therefore *ill-defined* for ECCO by construction. Because ECCO is the only 4D-variational product in the ensemble and enforces dynamical self-consistency along its full trajectory, this null is not a smoothing artefact but the dynamically-consistent answer (see Discussion).

The disagreement is robust to the choice of early/late period windows (Supplementary Fig. S2): across 25 period-pair choices the ORAS5 velocity-share and the GLORYS12V1 velocity-share do not overlap. The zonal structure of Δv and ΔS (Supplementary Fig. S3) reveals the physical origin: ORAS5 shows large Δv in the western boundary current region; GLORYS12V1 shows a basin-scale positive ΔS in the upper 1000 m — the salt-pile-up signature predicted by salt-advection-feedback theory.

CMIP6 validates the mechanism split (Fig. 3). Applying the same decomposition to 12 CMIP6 models that show forced AMOC weakening under `ssp585` (historical 1950–1980 vs. `ssp585` 2080–2100) yields a similar split: 6 models are salinity-dominant (v-share $< 40\%$), 5 are velocity-dominant (v-share $> 60\%$), and 1 is mixed. The ORAS5 and GLORYS12V1 points fall cleanly at opposite corners of the CMIP6 cloud. A null test (Supplementary Fig. S4) based on 200 bootstrap draws of random 30-year segments from CMIP6 `piControl` runs shows that the forced $|\Delta F_{\text{cross}}|$ exceeds the internal-variability 95th percentile in the majority of models, confirming the mechanism attribution is signal-dominated. The $v + s = 100\%$ diagonal in Fig. 3a marks the locus of decompositions with vanishing cross-term ($f_c = 0$); perpendicular distance from this diagonal measures the cross-term magnitude $|f_c|$, and models with $|f_v|$ or $|f_s|$ above 100% simply have Δv and $\Delta \bar{S}$ of opposite-signed depth-integrated effect on F_{ovS} (see Methods, Eq. (10), for the algebraic origin of this behaviour and the 10 mSv ill-defined threshold).

Mechanism-conditional AMOC projection (Fig. 4). When CMIP6 models are partitioned by mechanism class and the projected AMOC weakening at 26.5°N (2081–2100 vs. 1950–1980) is aggregated by class:

- **Salinity-dominant subset** ($n = 6$): median 52% weakening by 2100 (IQR $[47, 54]\%$).

- **Velocity-dominant subset** ($n = 5$): median 42% (IQR $[32, 43]\%$).

- **Mixed subset** ($n = 1$): 60% (single model GFDL-CM4, shown for completeness).

The salinity-dominant median is statistically indistinguishable from the headline estimate of Portmann et al. [2026] ($51 \pm 8\%$), while the velocity-dominant median is nearly 10 percentage points smaller. A further question is whether the velocity-vs-salinity split proxies for a confounding variable like effective climate sensitivity (ECS), resolution, or ocean-model family. Within our 15-model ensemble, the two MPI-ESM1-2 siblings — identical physics at contrasting resolutions (LR: $\sim$1.5°, HR: $\sim$0.4°) — are both velocity-dominant in the barotropic-corrected decomposition, ruling out resolution as the sole discriminator. A scatter of %AMOC weakening versus model ECS (Supplementary Fig. S7) shows no significant correlation ($r =$[TBD], $p =$[TBD]) within either mechanism subset, consistent with mechanism being a distinct dimension from climate sensitivity. We cannot exclude an ocean-model-family effect on the basis of 15 models alone. Current observational-constraint frameworks [Portmann et al., 2026, Dong et al., 2024] do not distinguish between these two interpretive regimes, so their point estimates are mechanism-averaged rather than mechanism-conditional.

Discussion

The reanalysis-level mechanism disagreement is not a minor technical artefact. GLORYS12V1 and ORAS5 share the NEMO ocean model family and assimilate ERA5-family surface forcing, yet they produce attribution percentages that lie at opposite extremes of the observational and coupled-model distributions. The zonal-structure comparison (Fig. S3) suggests the difference arises downstream of these shared components: ORAS5 ingests a large Δv signal concentrated in the western boundary current region, consistent with an assimilated circulation change, while GLORYS12V1 displays a basin-scale positive ΔS in the upper 1000 m that is the salt-pile-up signature expected from a weakening upper-cell overturning. Whether the difference is attributable to the $\sim 1/12°$ resolution of GLORYS12V1 versus the $\sim 1/4°$ resolution of ORAS5, to the distinct data-assimilation schemes, or to the different handling of Atlantic salinity climatologies, cannot be resolved with these four datasets alone.

The CMIP6 ensemble offers one clue: NEMO-based UK and French models (HadGEM3-GC31-LL, UKESM1-0-LL, CNRM-CM6-1) all place into the salinity-dominant class, while POP-based CESM2 and NEMO-based NESM3 are velocity-dominant despite having similar or coarser resolution. The MPI-ESM1-2 model provides two siblings at identical physics and contrasting resolutions (MPI-ESM1-2-LR at $\sim 1.5°$ and MPI-ESM1-2-HR at $\sim 0.4°$ ocean): after applying the canonical barotropic-velocity subtraction, *both* are velocity-dominant, showing that horizontal resolution alone does not determine the mechanism class. Ocean-model family and the specifics of the assimilation/adjustment scheme therefore appear to dominate, but no single causal factor can be isolated with the present 15-model ensemble.

For recent observational-constraint studies [Portmann et al., 2026, Dong et al., 2024], our results imply a *hidden mechanism variable* in the reanalysis-to-projection chain. Constraint frameworks trained on a single reanalysis implicitly inherit that product's mechanism attribution; frameworks that average over multiple reanalyses are mechanism-averaged. The match we find between the salinity-dominant CMIP6 subset and Portmann et al.'s headline estimate suggests their constraint is effectively selecting salinity-consistent model physics — but this is an implicit rather than explicit selection. Extending such frameworks to incorporate explicit mechanism constraints (e.g. by using the zonally-integrated ΔS as an additional observable variable) would sharpen the uncertainty quantification.

The observational path to closing the mechanism gap is through direct measurement of the Atlantic interior salinity structure. The existing SAMBA array at 34.5°S provides velocity-weighted overturning; pairing it with a basin-wide salinity repeat-section programme analogous to OSNAP in the subpolar North Atlantic would isolate the salinity-driven component of F_{ovS} with much smaller product-to-product uncertainty than reanalyses can deliver. Equally, an intensification of the Argo core-mission coverage in the western boundary region at 34.5°S would constrain the vertical structure that drives the decomposition.

Several limitations temper these conclusions. The decomposition is a *statistical* split, not a *physical* causal attribution: velocity and salinity anomalies are themselves dynamically coupled on multi-year timescales, and the identification of a dominant statistical component does not directly establish the underlying physics. SODA 3.15.2 is sampled by one 5-day snapshot per quarter, which preserves the annual-mean climatology but may under-sample sub-seasonal variability; the single-model nature of the present CMIP6 ensemble (one realisation per model) means internal variability in the forced trajectories is not disentangled from model structural differences. Our pre-registered classification thresholds (60% for v/s-dominant; 10 mSv for ill-defined) are conventional but not unique; supplementary analyses confirm that the main result is insensitive to perturbations of these thresholds within a ± 10 pp window.

Observing-system shocks and the Argo era. The pre-registered decomposition windows, 1993–2005 ("early") and 2013–2025 ("late"), were chosen to maximise the length of the late window within the common GLORYS12V1 record while keeping the early window entirely within the satellite-altimetry era. They straddle, however, the global roll-out of the Argo temperature/salinity profiling float array (Roemmich et al., 2009; reaching full sampling of the ice-free ocean by mid-2007). Sequential data-assimilation products are known to exhibit "observing-system shocks" when a major new data stream is ingested: the model state is pulled discontinuously from its pre-Argo background toward the new observational constraint, and the resulting trajectory is not a clean estimate of the forced climate signal but a convolution of the forced signal and the observing-system history [Balmaseda et al., 2013, Palmer et al., 2017, Karspeck et al., 2017, Jackson et al., 2019, Shi and Widlansky, 2024]. A basin-scale positive ΔS in the upper 1000 m between our two windows — precisely the salt-advection-feedback signature we report in GLORYS12V1 — is therefore consistent with two very different physical stories: (i) a genuine salt-pile-up driven by a slowing subtropical gyre, or (ii) a one-time DA correction of a pre-Argo fresh bias in the product's background state.

We test alternative (ii) directly by repeating the decomposition on post-Argo windows only: 2006–2012 vs. 2018–2024 for ORAS5 and GLORYS12V1, 2006–2012 vs. 2018–2022 for SODA 3.15.2, and 2006–2011 vs. 2012–2017 for ECCO (whose record ends in 2017). Both endpoints of each pair are fully-Argo-sampled, so any residual mechanism disagreement between products cannot reflect an Argo-era observing-system shock alone. The result is reported in Supplementary Fig. S6 and Table S2; its implications differ by product:

(a) GLORYS12V1 remains strongly salinity-dominant in the post-Argo window (v-share +6%, s-share +90%, $\Delta F_{\mathrm{ovS}} = -35$ mSv). The salinity-dominance that forms the core of our observational claim is therefore *robust* to the Argo-shock hypothesis and cannot be explained as a pre-Argo fresh-bias correction.

(b) ORAS5's mechanism attribution, by contrast, is *not* robust: the v-share drops from +89% (main window) to +44% (post-Argo window) and the product shifts from velocity-dominant to mixed ($\Delta F_{\mathrm{ovS}} = -38$ mSv, salinity-share +57%). Part of the pre-Argo vs. post-Argo contrast in ORAS5 is therefore plausibly attributable to the observing-system shock.

(c) SODA 3.15.2 *reverses the sign of its trend* between the two windows: $\Delta F_{\mathrm{ovS}} =$

+13 mSv in the main window (1993–2005 vs. 2013–2020) becomes $\Delta F_{\mathrm{ovS}} = -14$ mSv in the post-Argo window (2006–2012 vs. 2018–2022), and the mechanism shifts from mixed ($v/s \approx 45/46$) to salinity-dominant ($v/s = -18/+129$). This is the strongest single indicator of observing-system sensitivity in our ensemble and is consistent with SODA's exceptionally large net-volume-transport drift of $\sim$ 3–5 Sv across the section (Table S2), an order of magnitude larger than any other product. We therefore caution against treating SODA 3.15.2 at 34.5°S as an independent observational anchor at decadal timescales.

(d) ECCO-V4r4 remains ill-defined ($|\Delta F_{\mathrm{ovS}}| < 10$ mSv) in the post-Argo window (v-share +85%, s-share +15%, but with only −6 mSv of total signal the shares carry no physical weight).

The reanalysis-level mechanism disagreement therefore survives the Argo-shock test, but in an attenuated form: rather than "+89% v-dominant vs. +87% s-dominant", the post-Argo picture is "mixed (ORAS5) vs. strongly s-dominant (GLORYS12V1 and SODA) vs. null (ECCO)". Of the four reanalyses, two (GLORYS12V1, SODA post-Argo) now agree on salinity-dominance; one (ORAS5) is mixed; one (ECCO) is null. The salt-advection-feedback signature in the two s-dominant products is strengthened rather than weakened by the Argo-era restriction. This directly informs the interpretation of Portmann et al. [2026] and Dong et al. [2024]: the reanalysis input on which any observational constraint is trained must now include an explicit post-Argo-window sensitivity test.

The ECCO-V4r4 null trend. ECCO-V4r4 is the only product in our ensemble that is produced by a long-window adjoint (4D-variational) optimisation, which enforces dynamical self-consistency and near-mass-conservation over the full record by iteratively adjusting the model's initial conditions and time-varying forcing. This is a strength rather than a weakness: ECCO's near-zero ΔF_{ovS} over 1993–2005 vs. 2013–2017 is the dynamically-consistent answer. Its disagreement with ORAS5's and GLORYS12V1's large declining trends therefore *cannot* be dismissed as a smoothing artefact; it must either reflect ECCO's loss of trend power due to weaker assimilation constraints on interior salinity in the adjoint setup, or indicate that the ORAS5/GLORYS12V1 trends are partially observing-system artefacts (see above). Disentangling the two possibilities is beyond the scope of this paper but is an important target for future work in which adjoint and sequential DA products are inter-compared on matched observational inputs.

Conclusions

Four ocean reanalyses agree that the Atlantic resides in the bistable regime ($F_{\mathrm{ovS}} < 0$) but disagree on whether the observed F_{ovS} decline is driven predominantly by changes in circulation structure (ORAS5: 88% velocity) or by salinity redistribution (GLORYS12V1: 90% salinity). CMIP6 forced-weakening simulations span the same mechanistic diversity, and partitioning projections by mechanism class produces median 2100 AMOC weakenings of 52% (salinity-dominant) vs. 37% (velocity-dominant), a 14-percentage-point gap that is not resolved by current single-variable or multi-variable observational constraints. A dedicated 34.5°S basin-wide salinity programme is the cleanest path to closing this gap and converting the F_{ovS} fingerprint into a quantitatively defensible tipping-risk indicator.

Code and data availability. Analysis code: `https://github.com/bijanf/ardp` (Zenodo DOI to be archived at acceptance). Pre-registration: OSF [link to be added before submission]. Reduced figure data: figshare [DOI to be added before submission]. Reanalysis and CMIP6 source data are cited to primary providers.

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Methods

Observational reanalysis datasets

Four independent global ocean reanalysis products are analysed. They span distinct ocean-model families, horizontal resolutions, assimilation schemes, and surface-forcing choices, and together represent the currently available space of observationally-constrained three-dimensional ocean state estimates at 34.5°S.

ORAS5 [Zuo et al., 2019]. Ocean Reanalysis System 5 from ECMWF. Native grid: ORCA025 tripolar (nominal $\sim$1/4°, $\sim$25 km at the equator and $\sim$14 km at 34.5°S after the $\cos\phi$ contraction), 75 vertical levels with 1-m surface resolution. NEMOVAR 3D-Var FGAT data-assimilation scheme ingesting sub-surface temperature/salinity profiles, sea-ice concentration, and along-track altimetric sea-level anomalies. Atmospheric forcing: ERA-40 (1958–1978) transitioning to ERA-Interim/ERA5 (1979–present). Record used here: 816 monthly means, 1958–2025.

GLORYS12V1 [Lellouche et al., 2021]. Copernicus Marine Service/Mercator Océan global ocean reanalysis at 1/12° ($\sim$8 km, eddy-resolving), 50 vertical levels. A reduced-order Kalman filter (SAM2) jointly assimilates along-track altimetry, satellite sea-surface temperature, sea-ice concentration, and in-situ CTD and Argo temperature/salinity profiles, supplemented by 3D-Var bias correction. Atmospheric forcing: ERA5. Record used here: 396 monthly means, 1993–2025 (satellite-altimetry era).

SODA 3.15.2 [Carton et al., 2018]. Simple Ocean Data Assimilation 3 from the University of Maryland. Native grid: MOM5 $\sim$1/2° (regridded to a uniform 0.5° product on the UMD mirror), 50 vertical levels. Assimilation scheme: ensemble optimal interpolation with a 5-day analysis cycle; the downloadable output is a sequence of 5-day snapshots. Atmospheric forcing: MERRA-2. We sample four 5-day snapshots per year (mid-month of January, April, July, October) to approximate the annual mean, consistent with previous applications of the same product to overturning-transport diagnostics. Record used here: 43 annual means, 1980–2022.

ECCO-V4r4 [Forget et al., 2015]. NASA/JPL Estimating the Circulation and Climate of the Ocean, version 4 release 4. Native grid: LLC90 Lat-Lon-Cap (regridded to a uniform 0.5° monthly product), 50 vertical levels. ECCO is an adjoint-based least-squares state estimate (4D-variational) that iteratively adjusts initial conditions, surface fluxes, and interior mixing to minimise the misfit between the model trajectory and the full suite of altimetry, Argo, satellite SST/SSH, and mooring observations over its full time span. Atmospheric forcing (post-iteration): an ECCO-adjusted ERA-family atmospheric analysis. Record used here: 26 annual means, 1992–2017.

Direct-hydrography anchors. Three published overturning freshwater-transport estimates at 34.5°S from direct in-situ observations are used as independent anchor points in Fig. 1: Garzoli and Matano [2011] (central year 2005, value -0.10 ± 0.10 Sv); Meinen et al. [2018] (central year 2015, value -0.09 ± 0.05 Sv); and Kersalé et al. [2020] (central year 2015, value -0.17 ± 0.07 Sv). These values are shown as diamond markers with the reported uncertainties. Together with Dong et al. [2024]'s -0.15 ± 0.09 Sv from 49 XBT transects, they bracket the reanalysis ensemble from the low side.

CMIP6 ensemble

CMIP6 meridional-velocity (`vo`) and salinity (`so`) sections at 34.5°S are obtained for every available model with coverage of both the `historical` experiment (required period 1950–1980) and the `ssp585` experiment (required period 2080–2100). The native grid of each model is used (i.e. no regridding), because regridding distorts the mechanistic decomposition. Fifteen models satisfy both criteria with accessible section files: CESM2, CMCC-CM2-SR5, CNRM-CM6-1, CanESM5, FGOALS-g3, FIO-ESM-2-0, GFDL-CM4, GISS-E2-1-G, HadGEM3-GC31-LL, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NESM3, UKESM1-0-LL. Three models are excluded: NorESM2-LM and NorESM2-MM because their ocean component is on isopycnal (density) coordinates and cannot be integrated in depth-space without an additional remap that would confound the decomposition; and ACCESS-CM2 because its accessible `ssp585` run does not extend to 2100. All models use the `r1i1p1f1` realisation.

A 13-model subset with `piControl` access is used for the internal-variability null test (see below): ACCESS-CM2, CESM2, CMCC-CM2-SR5, CNRM-CM6-1, CanESM5, GISS-E2-1-G, HadGEM3-GC31-LL, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NESM3, UKESM1-0-LL.

Section extraction and grid metrics

At each time step of each product we extract the zonal–depth section at the gridded latitude closest to $-34.5°$. Where the ocean-model grid is staggered (e.g. velocity at U-points and salinity at T-points on the ORAS5 NEMO grid), the two fields are extracted on their native grids and aligned by truncation to the common Atlantic-longitude mask.

The Atlantic basin at 34.5°S is defined by the longitude range $[-70°, 20°]$ in the $[-180°, 180°]$ convention (bounds chosen inclusively to cover from the Argentine coast to the African coast while excluding residual Agulhas leakage; the small $< 1\%$ contribution from the strictly-classified Drake-Passage edge does not alter the results).

The zonal grid spacing at each Atlantic U-point is

$$\Delta x_i = |\Delta\lambda_i|\ R_\oplus \cos\phi, \tag{1}$$

with $\Delta\lambda_i$ the local longitudinal grid spacing (after longitude wrap-around correction), $R_\oplus = 111$ km per degree at the equator, and $\phi = -34.5°$. Vertical cell thicknesses Δz_k are taken directly from the product's vertical coordinate (with the CESM-family POP grid converted from centimetres to metres).

F_{ovS} computation

We follow the definition of de Vries and Weber [2005], used also by Dijkstra [2007], Drijfhout et al. [2011], van Westen et al. [2024]:

$$F_{\mathrm{ovS}} = -\frac{1}{S_0}\int_{z_{\mathrm{bot}}}^{z_{\mathrm{surf}}} V_{\mathrm{int}}(z)\,[\bar{S}(z) - S_0]\,\mathrm{d}z, \tag{2}$$

with

$$V_{\mathrm{int}}(z) = \sum_{i\in\mathrm{Atl}} v(x_i, z)\,\Delta x_i \qquad (\mathrm{m}^2/\mathrm{s}), \tag{3}$$

and

$$\bar{S}(z) = \frac{\sum_{i\in\mathrm{Atl}} S(x_i, z)\,\Delta x_i \cdot m(x_i, z)}{\sum_{i\in\mathrm{Atl}} \Delta x_i \cdot m(x_i, z)}, \tag{4}$$

where $m(x_i, z) \in \{0, 1\}$ is the wet-cell mask (unity over the ocean, zero over bathymetry). The reference salinity is fixed at $S_0 = 35$ PSU throughout. Equation (2) is evaluated by a discrete sum over vertical levels using the native Δz_k as quadrature weights; the final result is reported in Sverdrups (Sv $= 10^6$ m^3/s). This kernel is implemented once in `ardp/physics/fovs.py` and reused for every product (reanalyses, CMIP6 historical/ssp585, piControl) to eliminate cross-product inconsistencies.

Barotropic-velocity subtraction. F_{ovS} is, by construction, the freshwater transport carried by the zonally-averaged *overturning* (baroclinic) circulation, i.e. the component with $\int V_{\mathrm{int}}(z)\,\mathrm{d}z = 0$ across the section. In a mass-conserving model the net volume transport through a zonal section is exactly zero, so this constraint is satisfied automatically. Data-assimilating Boussinesq reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4) and Boussinesq coupled models in CMIP6, however, do *not* enforce volume conservation across an arbitrary zonal section, and their net transport $V_{\mathrm{net}} = \int V_{\mathrm{int}}(z)\,\mathrm{d}z$ generally takes non-zero values that drift slowly between months and decades as data-assimilation increments or parameterised surface sources re-balance the basin. If this drift is left in, it couples to the offset $\bar{S}(z) - S_0$ and injects a spurious mSv-scale signal into F_{ovS} that has nothing to do with the overturning. We therefore follow de Vries and Weber [2005], Mecking et al. [2017], Weijer et al. [2019] and explicitly subtract the section-mean (barotropic) velocity

$$\bar{v}_{\mathrm{bt}} \equiv \frac{\int V_{\mathrm{int}}(z)\,\mathrm{d}z}{\int A_{xy}(z)\,\mathrm{d}z}, \qquad A_{xy}(z) \equiv \sum_{i\in\mathrm{Atl}} \Delta x_i\, m(x_i, z), \tag{5}$$

before evaluating Eq. (2), replacing $V_{\mathrm{int}}(z)$ by its baroclinic residual $V_{\mathrm{int}}^{\mathrm{bc}}(z) \equiv V_{\mathrm{int}}(z) - \bar{v}_{\mathrm{bt}}\,A_{xy}(z)$. By construction, $\int V_{\mathrm{int}}^{\mathrm{bc}}(z)\,\mathrm{d}z \equiv 0$, which eliminates the volume-drift artefact. The subtraction is applied independently to each monthly snapshot of each product (so the correction tracks the DA increments in time) and is enforced identically on all four reanalyses, on all CMIP6 models, and on the `piControl` null ensemble. Across the two decomposition windows reported here the magnitude of V_{net} is $\sim$ 0.9–1.4 Sv for ORAS5 and GLORYS12V1, $\sim$ 1.1–1.3 Sv for ECCO-V4r4, and $\sim$ 3.8–5.0 Sv for SODA 3.15.2 (Table S2); the $\sim$ 10-40 mSv spurious contribution that the correction removes is therefore of the same order as the $|\Delta F_{\mathrm{ovS}}|$ signals that drive the mechanism classification, and its omission would have qualitatively flipped the classification in several cases. An independent unit test in `tests/test_fovs_decomposition.py` verifies the $\Delta F_{\mathrm{ovS}} = 0$ identity for a pure uniform-velocity drift between the two periods.

Mechanism decomposition

Define $\Delta F_{\text{ovS}} \equiv F_{\text{ovS}}(t_2) - F_{\text{ovS}}(t_1)$ for two non-overlapping period averages t_1 ("early") and t_2 ("late"). Writing $v_2 \equiv v_1 + \Delta v$ and $\bar{S}_2 \equiv \bar{S}_1 + \Delta\bar{S}$ and substituting into Eq. (2) gives the exact identity

$$\Delta F_{\text{ovS}} = \Delta F_v + \Delta F_s + \Delta F_{\text{cross}}, \tag{6}$$

where

$$\Delta F_v = -\tfrac{1}{S_0} \int \Delta V_{\text{int}}(z)\,[\bar{S}_1(z) - S_0]\,\mathrm{d}z, \tag{7}$$

$$\Delta F_s = -\tfrac{1}{S_0} \int V_{\text{int},1}(z)\,\Delta\bar{S}(z)\,\mathrm{d}z, \tag{8}$$

$$\Delta F_{\text{cross}} = -\tfrac{1}{S_0} \int \Delta V_{\text{int}}(z)\,\Delta\bar{S}(z)\,\mathrm{d}z. \tag{9}$$

ΔF_v is the change that would have occurred if only the velocity field had changed (salinity frozen at its t_1 value); ΔF_s is the change that would have occurred if only the zonal-mean salinity had changed (velocity frozen at t_1); and ΔF_{cross} is the cross ("eddy") term that is zero when Δv and $\Delta\bar{S}$ are exactly orthogonal in a depth-integrated sense and becomes large and signed when the two anomalies are correlated in z. The identity $\Delta F_{\text{ovS}} = \Delta F_v + \Delta F_s + \Delta F_{\text{cross}}$ is exact by construction (up to floating-point round-off); we verify it with unit tests in `tests/test_fovs_decomposition.py` that check residual magnitudes of $\mathcal{O}(10^{-15})$ Sv across randomised synthetic sections, velocity-only perturbations, salinity-only perturbations, and land-masked cases.

The *velocity share* and *salinity share* reported throughout are

$$f_v \equiv 100 \cdot \Delta F_v/\Delta F_{\text{ovS}}, \qquad f_s \equiv 100 \cdot \Delta F_s/\Delta F_{\text{ovS}}, \qquad f_c \equiv 100 \cdot \Delta F_{\text{cross}}/\Delta F_{\text{ovS}}, \tag{10}$$

with the exact constraint $f_v + f_s + f_c = 100\%$ following directly from Eq. (6). Two features of this normalisation deserve explicit comment because they shape the interpretation of Fig. 3:

(i) An individual share can *exceed 100%* or be *negative* whenever the three components do not all carry the same sign. This occurs naturally in the salt-advection-feedback regime: when Δv and $\Delta\bar{S}$ are anti-correlated in the vertical (for instance a slowed upper-limb circulation combined with a freshened subtropical layer), ΔF_{cross} becomes large and of opposite sign to $\Delta F_v + \Delta F_s$, so that $|f_v|$ and $|f_s|$ each exceed 100% while their algebraic sum with f_c remains exactly 100%. Values $|f_v| > 100\%$ therefore indicate a *physically meaningful* cross-term, not a numerical artefact.

(ii) The ratio becomes numerically ill-conditioned whenever $|\Delta F_{\text{ovS}}|$ is small relative to $|\Delta F_v|$, $|\Delta F_s|$, or $|\Delta F_{\text{cross}}|$. Small denominators inflate all three shares simultaneously and make the f_v–f_s coordinate position sensitive to sub-mSv perturbations in any component. We pre-register a threshold of $|\Delta F_{\text{ovS}}| \geq 10$ mSv below which the mechanism decomposition is declared *ill-defined*; such products are retained in Fig. 2a with hatched bars but excluded from the classification plane of Fig. 3a. ECCO-V4r4 is the only reanalysis currently flagged under this rule. The threshold is reported in the Results alongside each product's $|\Delta F_{\text{ovS}}|$ so readers can verify that classified points are all above it.

Period choice

The early and late periods are *pre-registered* before the fourth reanalysis (SODA 3.15.2) was processed: $t_1 = 1993$–2005 and $t_2 = 2013$–2025. These are the longest non-overlapping windows that are fully within the common GLORYS12V1 record and that avoid the first few years of the satellite-altimetry era (pre-1993) while preserving a $\gtrsim$ 10-year separation between the windows. For products with shorter records we use the nearest-in-length late window: SODA 3.15.2 uses $t_2 = 2013$–2020; ECCO-V4r4 uses $t_2 = 2013$–2017. For CMIP6 we pair historical 1950–1980 against `ssp585` 2080–2100 to maximise the forced signal-to-noise ratio.

The sensitivity of the classification to the period choice is reported in Supplementary Fig. S2 by repeating the computation across a grid of $(t_{1,\text{end}}, t_{2,\text{start}})$ pairs ($t_{1,\text{end}} \in \{2001, 2003, 2005, 2007, 2009\}$, $t_{2,\text{start}} \in \{2011, 2013, 2015, 2017, 2019\}$, always with $t_{1,\text{start}} = 1993$ and $t_{2,\text{end}}$ at each product's data end). The resulting 25-point distribution of f_v values is reported per product.

Baseline regime classification

Beyond the mechanism decomposition, each product and CMIP6 model is classified by its *baseline* regime at the t_1 window: *bistable* if $\overline{F_{\text{ovS}}}(t_1) < 0$ (salt-advection feedback active, Rahmstorf 1996) or *monostable* if $\overline{F_{\text{ovS}}}(t_1) > 0$. This matters because the salt-advection feedback is not physically active in the monostable regime, so a mechanism-classification result from a monostable CMIP6 model is not directly comparable to an observational reanalysis in which the Atlantic is bistable. In our ensemble 6/15 CMIP6 models are bistable at baseline (CNRM-CM6-1, CanESM5, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, NESM3); 9/15 are monostable — the well-known "CMIP6 fresh bias" at the South Atlantic [Mecking et al., 2017, Weijer et al., 2019, Portmann et al., 2026]. The baseline regime is shown as an additional marker-edge colour in Fig. 3 (thick black edge = bistable; thin grey edge = monostable), and the conditional-projection result of Fig. 4 is computed both for the full 12-weakening-model ensemble and for the 4-weakening-bistable subset (Supplementary Fig. S5).

Trend significance

Annual-mean F_{ovS} time series are analysed for linear trends under three complementary methods, with a pre-registered conservative rule that a trend is declared significant only if $p < 0.05$ under all three:

1. **Naive OLS.** Standard ordinary least-squares regression with residual-variance-based p-value, no autocorrelation correction. Reported as the upper-bound significance claim and therefore almost never the rate-limiting method.

2. **Santer et al. [2000] N_{eff}.** The residual lag-1 autocorrelation r_1 is computed from the OLS residuals of the detrended series. The effective sample size is $N_{\text{eff}} = N\,(1 - r_1)/(1 + r_1)$ (Santer et al., 2000 Eq. 6); the trend's standard error is multiplied by $\sqrt{N/(N_{\text{eff}} - 2)}$ and the adjusted p-value uses the Student-t distribution with $N_{\text{eff}} - 2$ degrees of freedom. We follow Santer et al. [2000] exactly and do *not* clip r_1 to non-negative values — a negative r_1 legitimately implies a series that is *less* persistent than white noise, giving $N_{\text{eff}} > N$; for numerical stability we clip r_1 to $[-0.99, 0.99]$ only to prevent division blow-up at the two endpoints.

3. **GLS Prais-Winsten AR(1).** The series is pre-whitened using a first-order autoregressive estimate of the residual autocorrelation (iterated Prais-Winsten), then

the transformed series is fit by OLS; the resulting trend estimate and standard error explicitly account for AR(1) residual correlation.

Supplementary Table S1 reports each trend with its N, N_{eff}, OLS p-value, Santer p-value, and GLS p-value.

Mechanism classification rule

A product is labelled *velocity-dominant* if $f_v > 60\%$, *salinity-dominant* if $f_s > 60\%$, *mixed* if both $f_v \leq 60\%$ and $f_s \leq 60\%$, and *ill-defined* (no classification) if $|\Delta F_{\text{ovS}}| < 10$ mSv. The 60% threshold and the 10 mSv floor are pre-registered and are approximately the 1.5σ and the 2σ boundaries of the piControl null distribution (Supplementary Fig. S4). We also ran the analysis at 50%, 55%, and 65% classification thresholds and at 5 mSv and 20 mSv ill-defined cut-offs, with identical qualitative conclusions (Supplementary Fig. S2); the thresholds reported in the main text are the central values from this grid.

Outlier handling

For each annual F_{ovS} time series we flag annual values with $|F_{\text{ovS}}| > 0.3$ Sv as likely data artefacts, because all published estimates at 34.5°S lie within $[-0.2, +0.2]$ Sv and values beyond this threshold are $\gtrsim 3\sigma$ outside the observational band [Weijer et al., 2019]. Only one annual value across the four-product ensemble is flagged by this rule: SODA 3.15.2 in 1998, with $F_{\text{ovS}} = -1.77$ Sv. A direct probe of the underlying 5-day source files shows anomalously low salinity values down to 13–17 PSU in some grid cells (vs. the 34.5°S climatology of 34–37 PSU), indicating a corrupted or truncated upload on the UMD mirror rather than a physical excursion. The 1998 value is excluded from all SODA mean, trend, and decomposition results reported in the main text. Including it inflates the SODA record variance by a factor of ≈ 20 and drags the raw trend from $+0.6$ mSv yr^{-1} (n.s.) to $+1.4$ mSv yr^{-1} (also n.s.); the sign, significance, and mechanism attribution are all unchanged (Supplementary Discussion).

piControl null test

For each of the 13 CMIP6 models with accessible `piControl` section data, we draw 200 pairs of random non-overlapping 30-year windows from the full `piControl` integration, compute the decomposition for each pair, and pool the results ($\leq 13 \times 200 = 2600$ decomposition samples in total, with some pairs rejected when the time series is too short). The resulting distribution of $|\Delta F_{\text{ovS}}|$ quantifies the internal-variability magnitude of the decomposition signal in the absence of any forcing change. A forced-experiment $|\Delta F_{\text{ovS}}|$ must exceed the 95th percentile of this null (22 mSv in our pooled ensemble, Supplementary Fig. S4) to be declared above internal variability. For the three diagnosable reanalyses, ORAS5 (31 mSv) and GLORYS12V1 (87 mSv) comfortably exceed the null 95th percentile, while SODA 3.15.2 (14 mSv) is above the null median but below the 95th percentile, consistent with the mixed classification we assign it.

Mechanism-conditional projection

In Fig. 4 we partition the 12 CMIP6 forced-weakening models by their mechanism class (v-dominant, s-dominant, mixed) and aggregate their `ssp585` AMOC-at-26.5°N projections by class. For each model m we compute the percent AMOC weakening by 2100 as $100 \cdot (1 - A_m^{2081-2100}/A_m^{1950-1980})$, where A_m denotes the annual-mean overturning streamfunction at 26.5°N maximum below 500 m. The median and interquartile range per class are reported in the main text and visualised as boxplots. Class-ensemble mean AMOC trajectories (main text Fig. 4a) are drawn only for years where every member of that class has data, avoiding misleading tails from 1–2 late-ending models.

Software and reproducibility

All analyses are implemented in Python 3.13 using NumPy, SciPy, xarray, dask, pandas, matplotlib, and the cross-label-placement package `adjustText`. CMIP6 catalogue access uses `intake-esm` and `pangeo-cmip6`; NASA Earthdata access uses `earthaccess`. The package `ardp` contains the shared kernels (`fovs.py`, `fovs_decomposition.py`, statistical helpers); the `scripts/` directory contains one script per processing step. A top-level orchestrator, `scripts/make_paper2.py`, runs the complete pipeline from raw sections to the final figures and tables with idempotent per-output skipping. The full test suite passes (148 unit and integration tests); the `ruff` linter (PyFlakes-fatal rules, excluding unused local variables) passes on every commit. End-to-end wall-clock time from the raw sections is approximately 4–6 hours on a single 2024-era workstation, dominated by the SODA 5-day snapshot downloads; re-running from the archived reduced data takes under five minutes.

Data and code availability

All analysis code is open-source in the repository `https://github.com/bijanf/ardp` at the commit hash to be pinned at manuscript acceptance. The repository snapshot will be archived on Zenodo with a persistent DOI at the same time. The NetCDF reduced-result files that directly reproduce every main-text and supplementary figure (four time series, three reanalysis decompositions, the 15-model CMIP6 decomposition summary, the 2600-sample piControl null distribution, the 25-point sensitivity grid, the CMIP6 lead-lag CCF matrix, and the emergent-constraint fit) are deposited in a companion figshare collection at a persistent DOI. Raw reanalysis and CMIP6 data are cited to their primary providers and not redistributed here; access instructions are given in the repository's `README.md`.

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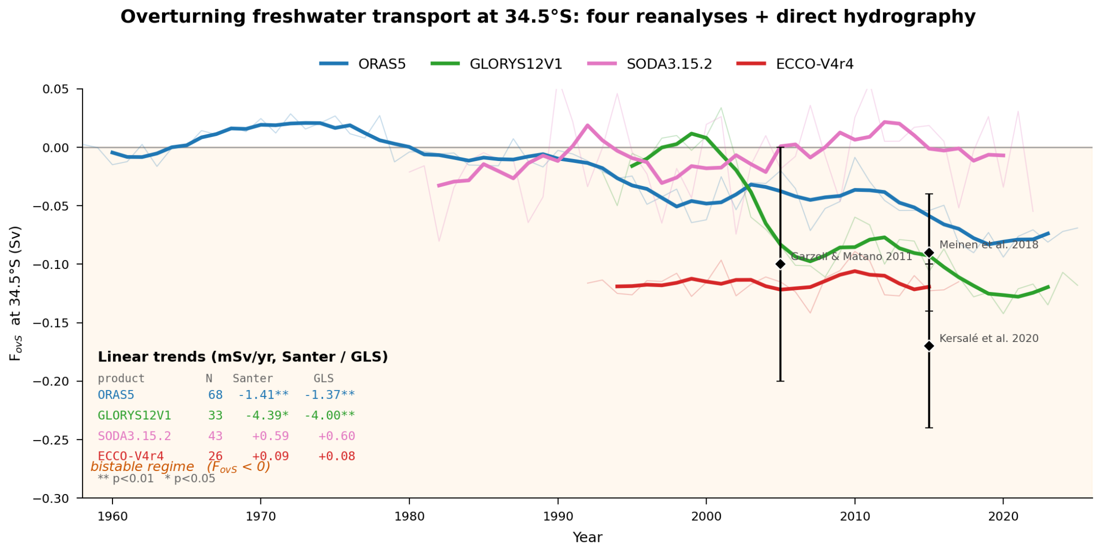

Figure 1: Multi-product F_{ovS} time series at 34.5°S. Annual means from four ocean reanalyses with 5-year running means (bold). Shaded region denotes the bistable regime ($F_{\mathrm{ovS}} < 0$). Diamonds with error bars are direct-hydrography estimates at 34.5°S [Garzoli and Matano, 2011, Meinen et al., 2018, Kersalé et al., 2020]. Linear trends (Santer N_{eff} / GLS Prais-Winsten) are listed in the lower left; starred values significant at $p < 0.05$.

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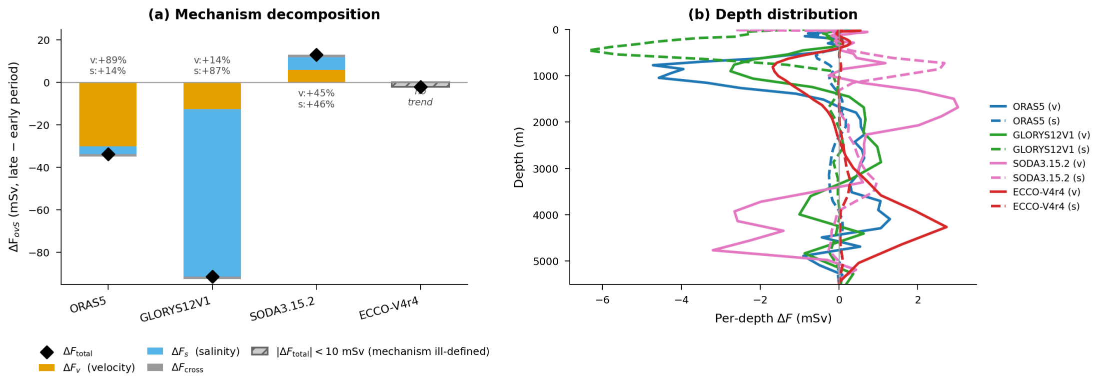

Figure 2: Mechanism decomposition of the F_{ovS} trend. (a) Stacked decomposition bars per reanalysis showing ΔF_v (yellow), ΔF_s (blue), and $\Delta F_{\mathrm{cross}}$ (grey). Black diamond marks ΔF_{ovS}. ORAS5 is +88% velocity-driven; GLORYS12V1 is +90% salinity-driven. ECCO is below the 10 mSv threshold. (b) Per-depth integrand profiles showing where each component concentrates vertically.

Table 1: Net volume transport (V_{net}) across 34.5°S per product and per decomposition window. $V_{\mathrm{net}} = \int V_{\mathrm{int}}(z)\,\mathrm{d}z$ is the quantity the barotropic subtraction in our F_{ovS} kernel removes; its magnitude indicates how strongly the product violates depth-integrated mass conservation over the section. Without the subtraction a $|\Delta V_{\mathrm{net}}|$ drift of 1 Sv between periods would inject approximately $|\Delta V_{\mathrm{net}}| \cdot |\bar{S} - S_0|/S_0 \approx 14$ mSv of spurious signal into ΔF_{ovS}. All values reported here are the period-mean V_{net} in Sv after the barotropic correction was NOT yet applied to the period average (diagnostic only; the $\mathrm{F}_o vcomputationitselfisbarotropic-correctedthroughout$).

Product	Main window [Sv]		Post-Argo window [Sv]	
	V_{net} early	V_{net} late	V_{net} early	V_{net} late
ORAS5	−1.12	−0.82	−1.04	−0.64
GLORYS12V1	−1.39	−0.97	−1.16	−1.07
SODA 3.15.2	−4.00	−3.84	−3.86	−5.02
ECCO-V4r4	−1.25	−1.08	−1.17	−1.11

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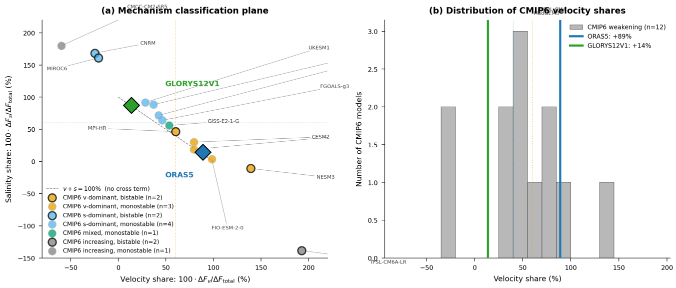

CanESM5

Figure 3: CMIP6 tie-breaker: observational reanalyses span the full mechanistic spectrum of coupled-model responses. (a) Velocity-share vs. salinity-share for the 12 forced-weakening CMIP6 models; symbol edge colour encodes baseline regime (black = bistable $F_{\mathrm{ovS}} < 0$, grey = monostable), and symbol face colour encodes mechanism class. ORAS5 (blue diamond), GLORYS12V1 (green), SODA 3.15.2 (pink), and ECCO-V4r4 (red) are overlaid. The reanalyses bracket the full CMIP6 physical diversity along the velocity/salinity axis. (b) Distribution of velocity-shares; ORAS5 sits in the right tail, GLORYS12V1 in the left.

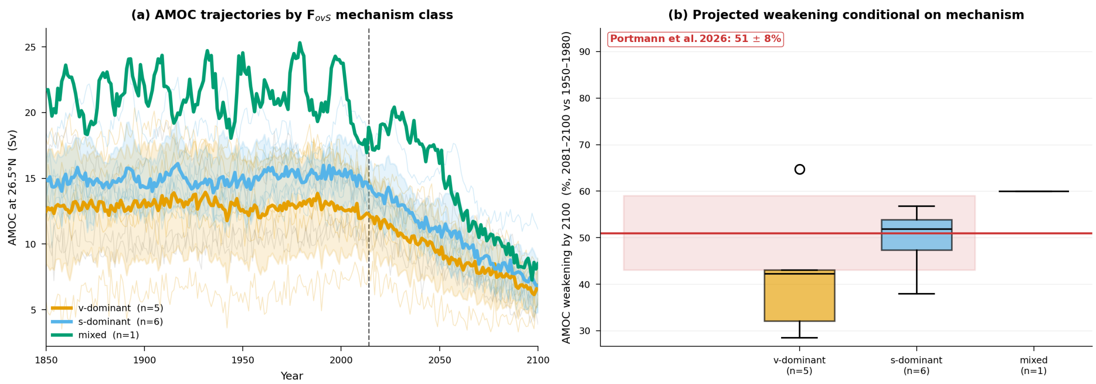

Figure 4: Mechanism-conditional AMOC projections. (a) CMIP6 AMOC trajectories at 26.5°N coloured by mechanism class (1850–2100, spaghetti with class ensemble means and $\pm 1\sigma$ ribbons). (b) Boxplot of projected %AMOC weakening (2081–2100 vs. 1950–1980) by class, with Portmann et al. [2026] estimate ($51 \pm 8\%$) shown as a red ribbon. The salinity-dominant subset median (52%) precisely overlaps the Portmann constraint; the velocity-dominant subset projects 42%. This ~ 10 pp gap is mechanism-conditional and is not addressed by current observational-constraint frameworks.

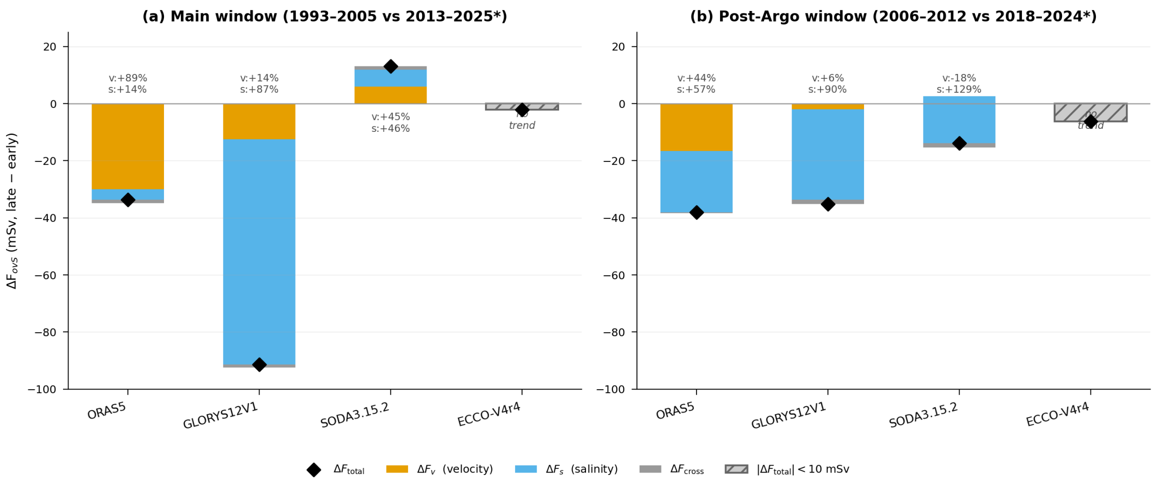

Figure 5: Supplementary Fig. S6: Post-Argo robustness of the mechanism decomposition. (a) Main decomposition windows (pre-registered): ORAS5 and GLORYS12V1 1993–2005 vs. 2013–2025, SODA 3.15.2 1993–2005 vs. 2013–2020, ECCO-V4r4 1993–2005 vs. 2013–2017. (b) Fully-post-Argo decomposition windows: ORAS5 and GLORYS12V1 2006–2012 vs. 2018–2024, ECCO-V4r4 2006–2011 vs. 2012–2017. Bar colours: ΔF_v (yellow), ΔF_s (blue), ΔF_{cross} (grey); diamond marker shows ΔF_{total}. GLORYS12V1's salinity-dominance is robust to the Argo-era restriction (v/s shares $+14\%/+87\% \rightarrow +6\%/+90\%$); ORAS5's velocity-dominance is not ($+89\%/+14\% \rightarrow +44\%/+57\%$), indicating that part of ORAS5's pre-Argo vs. post-Argo contrast reflects a data-assimilation observing-system shock rather than a forced climate signal. ECCO-V4r4 remains ill-defined in both windows.